MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
49R0 INTERNAL ROADS AND DRAINAGE : ACCESS ROAD
DESIGN CALCULATION

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DEFINITIONS AND ABBREVIATIONS

Document	;	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	;	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	;	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	;	Ma'aden Project Management Team
'MMGP Project'	;	Mansourah Massarah Gold Processing Project
Attribute	;	Information element that describes a document or tag.
Supplier	;	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	;	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	;	Quality Assurance
QC	;	Quality Control
Works	;	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The document is prepared for the design calculation of the Access Road to Mansourah Massarah Process Plant.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The design document is prepared for the Access Road to Mansourah Massarah Process Plant.
The document is prepared based on the following input documents :

a) MD-216-4300-EG-CV-CRT-0001 (Rev C01)_ Roads Design Criteria

The document is also prepared based on the following approved document:
a) MD-216-49RA-EG-CV-CAL-1637 A02

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4. TRAFFIC DATA

Vehicle Type	Total Vehicle Load (t)	Single Axle load (t)	Single Axle load (kip)	No. of passes per day
5AX Tendem-Tendem	49.5	12	26.4	125

NB. Single Axle load, No. of passes (in above table) are reasonably assumed.

5. DESIGN PARAMETERS

CBR					
CBR value of subgrade		=	25 %		(MD-216-49RA-EG-CV-RPT-1001-A01 (Geotechnical Investigation Report Plant Access Road))
Analysis Period					
Analysis period for road design		=	20 years		(Cl 5.2, MD-216-4300-EG-CV-CRT-0001 (Roads Design Criteria))
Reliability					
Reliability	R	=	65 %		(Cl 2.1.3, Table 2.2,Part II,AASHTO)
Standard normal deviate	Z_R	=	-0.3885		(Table 4.1, Part I, AASHTO)
Serviceability loss					(Cl 2.2.1, Part II, AASHTO)
Initial Serviceability Index	p_o	=	4.2		
Terminal Serviceability Index	p_t	=	2.5		
Serviceability loss, ΔPSI	$p_o - p_t$	=	1.7		
Standard deviation					(Cl 2.1.3, Part II, AASHTO)
Standard deviation error	S_0	=	0.45		
Pavement layer materials characterization					(Cl 2.3.3, Part II, AASHTO)
Effective resilient modulus of -					
Asphalt Concrete	E_{AC}	=	400000	psi	
Granular Base	E_{BS}	=	30000	psi	
Granular Subbase	E_{SB}	=	15000	psi	
Roadbed material	M_R	=	12500	psi	
Layer Coefficients					
Asphalt Concrete	a_1	=	0.42		(Fig 2.5, Cl. 2.3.5, AASHTO)
Granular Base	a_2	=	0.14		(Fig 2.6, Cl. 2.3.5, AASHTO)
Granular Subbase	a_3	=	0.11		(Fig 2.7, Cl. 2.3.5, AASHTO)
Drainage coefficient					(Cl 2.4.1 and Table 2.4, AASHTO)
Base material	m_2	=	1.3		
Subbase material	m_3	=	1.3		

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6. FUTURE TRAFFIC ESTIMATION

Estimated future traffic	w_{18}	=	D_D x D_L x $\hat{w}_{18}$	(Cl 2.1.2, Part II, AASHTO)
Directional distribution factor	D_D	=	0.5	
Lane distribution factor	D_L	=	100 %	
Traffic growth rate per year		=	1 %	(Assumed)
Traffic growth factor		=	22.15	(Table D.20, AASHTO)
ESAL Factor		=	5.93	(Table D.5, AASHTO)

Cumulative two-directional 18-kip ESAL units during the analysis period ($\hat{w}18$) shall be determined as shown below:

Design traffic (per year)	=	Current traffic x Traffic growth factor x 365
	=	1010594
Cumulative Equivalent Single Axle Load (ESAL)	=	Design traffic (per year) x ESAL factor
	=	5992821
Estimated future traffic	=	D_D x D_L x $\hat{w}_{18}$
	=	2996410
	=	3.00 x 10^6 ESAL

7. STRUCTURAL NUMBER CALCULATION

(Cl 3.1.1, Part II, AASHTO)

Structural number shall be calculated by following equation :-

$$\log_{10}w_{18} = z_R S_0 + 9.36 \log_{10}(SN+1) - 0.20 + \{\log_{10}[\Delta PSI/(4.2-1.5)] / (0.40+1094/(SN+1)^{5.19})\} + 2.32 \log_{10}M_R - 8.07$$

where

w_{18}	Estimated future traffic
Z_R	Standard normal deviate
SN	Structural number
ΔPSI	Serviceability loss
M_R	Effective resilient modulus of roadbed material

$z_R S_0$	=	a	=	-0.174825
9.36 $\log_{10}(SN+1)$ - 0.20	=	b	=	see below table
$\log_{10}[\Delta PSI/(4.2-1.5)]$	=	c	=	-0.200915
$0.40+1094/(SN+1)^{5.19}$	=	d	=	see below table
2.32 $\log_{10}M_R$ - 8.07	=	e	=	1.4348312

Trial	Stability number (SN)	b	d	$\log_{10}w_{18}$	w_{18}
1	3.000	5.43528	1.22097	6.5307339	3394173
2	2.900	5.33236	1.33625	6.4420138	2767029
3	2.950	5.38415	1.27635	6.4867416	3067197
4	2.945	5.37900	1.28213	6.4823025	3036005
5	2.940	5.37384	1.28796	6.4778559	3005079

Calculated Structural Number SN_{cal} = 2.940

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8. SELECTION OF LAYER THICKNESSES

The following equation provides basis for converting SN into actual thicknesses of surfacing, base and subbase. The SN equation does not have a single unique solution; i.e., there are many combinations of layer thicknesses that are satisfactory solutions. The thickness of the flexible pavement layers is rounded to the nearest next 1/2 inch.

SN = $a_1D_1 + a_2D_2m_2 + a_3D_3m_3$

where

a_1, a_2, a_3 = Layer coefficients of surface, base and subbase course respectively

D_1, D_2, D_3 = Actual thickness of surface, base and subbase course respectively

m_2, m_3 = Drainage coefficients of base and subbase course respectively

a_1	a_2	a_3	D_1	D_2	D_3	m_2	m_3	$SN_{provided}$
0.420	0.140	0.110	3	5.9	11.8	1.3	1.3	4.021

Since $SN_{provided}$ > SN_{cal} OK

Hence provide following layer thicknesses -

Surface course	=	3.0	inch	=	80	mm
Granular Base Course	=	5.9	inch	=	150	mm
Granular Subbase Course	=	11.8	inch	=	300	mm

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